

PYTHON PROGRAMMING

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INTRODUCTION TO PYTHON UNIT -1

Introduction :

- Features and Applications of Python;
- Python Versions;
- Installation of Python;
- Python Command Line mode and Python IDEs;
- Simple Python Program.

Python Basics:

- Identifiers;
- Keywords;
- Statements and Expressions;
- Variables;
- Operators;
- Precedence and Association;
- Data Types;
- Indentation;

- Comments;
- Built-in Functions- Console Input and Console Output,
- Type Conversions;
- Python Libraries;
- Importing Libraries with Examples.
- Python Control Flow:
- **Types of Control Flow:**
 - ❖ Control Flow Statements- if, else, elif, while loop,
 - ❖ break, continue statements,
 - ❖ for loop Statement;
 - ❖ range () and exit () functions

○ PYTHON PROGRAMMING

Very beginning learn about

What?

Where?

Why?

○ What is Python?

- Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.
- OR
- Python is an interpreted, object-oriented, high-level programming language with dynamic semantics developed by Guido van Rossum.
- It was originally released in 1991. Designed to be easy as well as fun, the name "Python" is a nod to the British comedy group Monty Python.
- Python has a reputation as a beginner-friendly language, replacing Java as the most widely used introductory language because it handles much of the complexity for the user, allowing beginners to focus on fully grasping programming concepts rather than minute details.

○ Where It is used for?

- web development (server-side),
- software development,
- mathematics,
- system scripting.
- Processing [big data](#)
- Fast prototyping
- Developing production-ready software

○ Why Python?

- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Things to know

- The most recent major version of Python is Python 3, which we shall be using .
- However, Python 2, although not being updated with anything other than security updates, is still quite popular.
- Python will be written in a text editor.
- It is possible to write Python in an Integrated Development Environment, such as Thonny, Pycharm, Netbeans or Eclipse which are particularly useful when managing larger collections of Python files.



- Example
- `print("Hello, World!")`
- **Python Install**
- Many PCs and Macs will have python already installed.
- To check if you have python installed on a Windows PC, search in the start bar for Python or run the following on the Command Line (cmd.exe):

```
C:\Users\Your Name>python -version
```

If you find that you do not have Python installed on your computer, then you can download it for free from the following website: <https://www.python.org/>



○ **Verify installation**

- Once IDLE is launched, go to →file menu
- Click on → new file to create new python file.

○ **How to use IDLE**

- Print("morning")
- Save the file with .py extension and run the file(run menu→ click on run module or press F5)
- If the output is morning then installation is successful and IDLE is ready to use.

○ **Python Quickstart**

- Python is an interpreted programming language, this means that as a developer you write Python (.py) files in a text editor and then put those files into the python interpreter to be executed.
- The way to run a python file is like this on the command line:
- C:\Users\panka>AppData\Local\Programs\Python\Python311\my.py
- Where "my.py" is the name of your python file.
- Let's write our first Python file, called my.py, which can be done in any text editor.



- **The Python Command Line**

- To test a short amount of code in python sometimes it is quickest and easiest not to write the code in a file. This is made possible because Python can be run as a command line itself.
- Type the following on the Windows, Mac or Linux command line:
- **C:\Users\Your Name>python**
- Or, if the "python" command did not work, you can try "py":
- **C:\Users\Your Name>py**
- From there you can write any python, including our hello world example from earlier in the tutorial:
- **>>> print("Hello, World!")**
- Which will write "Hello, World!" in the command line:
- **Hello, World!**
- Whenever you are done in the python command line, you can simply type the following to quit the python command line interface:
- **>>>exit()**



- **Features and Benefits of Python**

- Compatible with a variety of platforms including Windows, Mac, Linux, Raspberry Pi, and others
- Uses a simple syntax comparable to the English language that lets developers use fewer lines than other programming languages
- Operates on an interpreter system that allows code to be executed immediately, fast-tracking prototyping
- Can be handled in a procedural, object-orientated, or functional way

- **Python Identifiers**

- Identifiers are the name given to variables, classes, methods, functions etc. For example

```
language = 'Python'
```

- Here, language is a variable (an identifier) which holds the value 'Python'.
- We cannot use keywords as variable names as they are reserved names that are built-in to Python. For example,

```
continue = 'Python'
```

- The above code is wrong because we have used continue as a variable name. (CONTINUE IS KEYWORD)